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Jeffrey-Bayes Dynamics as Natural Gradient Descent by a Recurrent Network

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Abstract

Combining Bayes' rule with Jeffrey-style probability kinematics yields a belief update algorithm with some interesting neural and geometric properties. It can be implemented as a recurrent neural network, with fixed weights encoding probabilistic domain knowledge. An initial probability vector encoding the system's current belief is propagated forward through the network, combined with an evidence vector, and then propagated backward to compute a multiplicative update to the current belief. By iterating this back-and-forth process, we obtain a dynamical system that generates a flow on the probability simplex of belief states. We use elementary information geometry to show that this flow is an example of natural gradient descent in Kullback-Leibler divergence. The neural implementation of this system is comparable to recent models of Bayesian computation in recurrent cortical circuits.

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1 Introduction

The proper update of probabilistic beliefs in light of uncertain evidence has been a continuing area of interest in statistics, artificial intelligence, and philosophy of science for at least four decades. Whereas Bayes' theorem gives a simple way to update a probability distribution given *certain* evidence, the use of uncertain evidence is more controversial. *Probability kinematics* (Jeffrey 1965; Diaconis and Zabell 1982; Chan and Darwiche 2005) is a deeply studied approach based on the redistribution of probabilities. While probability kinematics is one technique where the caveats may in fact outnumber the applications (see e.g. Grünwald and Halpern 2003), its longtime role as a natural default mode of reasoning is unassailable.

In this paper we link an extension of probability kinematics with neural computation and information geometry. The probability update method itself was explored in the context of pattern classification by Clausing (1993). The present study clarifies the nature of this method as a dynamical system, deriving it in a different fashion, presenting it as a connectionist computation, and showing that it, rather surprisingly, performs natural gradient descent (Amari 1998) in Kullback-Leibler divergence. This provides a pleasantly simple illustration of information geometry. Other examples of information geometry in layered recurrent networks, such as Helmholtz machines (Dayan et al. 1995), involve the adjustment of models using versions of the EM algorithm. In the present case, the model (implemented as connectionist weights) stays fixed; it is *evidence* that drives a flow through belief space. This is probabilistic inference, rather than parameter learning.

2 Jeffrey-Bayes Updating

Consider an exhaustive set of mutually exclusive hypotheses $\{h_1, h_2, \dots, h_n\}$, viewed as values of a random variable H . A *belief* is a probability assignment on this set. We regard it as a column vector $\mathbf{p} = [p(h_1) \ p(h_2) \ \dots \ p(h_n)]^T \equiv [p_1 \ p_2 \ \dots \ p_n]^T$, with $\sum_k p_k = 1$ and $0 < p_k < 1$. Let Δ^{n-1} denote the $(n-1)$ -dimensional space of such probability vectors, a probability simplex we call *belief space*. A belief update method is a discrete-time dynamical system on Δ^{n-1} . From an initial belief $\mathbf{p}(0)$ it will generate a trajectory $\mathbf{p}(t)$, $t = 1, 2, \dots$. The dynamics is driven by an external stream of evidence, conveyed by an attribute A . Let $\{a_1, a_2, \dots, a_m\}$ be an exhaustive list of the possible values of A . A fixed “knowledge base” of background assumptions is coded in the form $w_{ik} = p(a_i | h_k)$. These are the probabilities that the attribute A takes on the value a_i given that hypothesis h_k is true. These attribute values are exhaustive, so $\sum_i w_{ik} = 1$.

We treat these probabilities as entries in a matrix $\mathbf{W}$ that is constant in time. At time t , the system performs an observation, accumulating evidence $E(t)$, accessible only through a probability assignment to each of the attribute values. We write this as a vector $\mathbf{e}(t) = [e_1(t) \ e_2(t) \ \dots \ e_m(t)]^T \in \Delta^{m-1}$ with $e_i(t) = p(a_i | E(1), \dots, E(t))$. The probability simplex consisting of such vectors $\mathbf{e}(t)$ we will call *observation space*. The update method Φ described below updates $\mathbf{p}(t)$ given fixed knowledge $\mathbf{W}$ and current evidence $\mathbf{e}(t+1)$:

$$\mathbf{p}(t+1) = \Phi[\mathbf{W}, \mathbf{p}(t), \mathbf{e}(t+1)] .$$

Because $E(t)$ can be viewed as a kind of virtual evidence (Pearl 1988), this suggests a graphical model of the form $H \leftarrow A \rightarrow E(t)$. Here $E(t)$ is an abbreviation for a set of time-indexed nodes $\{E(1), E(2), \dots\}$ with possibly a high degree of conditional dependence between them. We will assume that H is independent of each $E(t)$ given A , as implied by this tree. Nevertheless, the algorithm we will study here is not equivalent to belief propagation in this model, and in fact the conditional probability tables that we are given are not the ones that would have been natural for that purpose.

To make the role of the probabilities in this dynamical system more clear, one can tell a brief story. An investigator tries to determine the type of explosive that caused a certain distribution of shrapnel. In this case, $p_k = p(h_k)$ is the investigator's degree of belief that an explosive of type k was the cause. Background knowledge (presumed constant during this investigation) is embodied in the matrix elements $w_{ik} = p(a_i | h_k)$, the probability that a shrapnel fragment from explosive type k has iron content at level a_i . Evidence is given by $e_i(t)$, the probability that samples collected during an observation at time t have iron content at level a_i . Note that the probabilities underlying $\mathbf{p}$ are inherently Bayesian, whereas those underlying $\mathbf{W}$ and $\mathbf{e}$ can admit a frequentist interpretation. Evidence really is virtual here; insofar as $E(t)$ represents an event, it is the event of an observation occurring. We don't know anything about the probabilities of these observation events, and the investigator only deals with the resulting attribute values. As such, the hypothesis (H) is independent of the evidence (E) given the value of the observed attributes (A).

The particular belief updating method we will investigate here involves a combination of Bayes' rule and Jeffrey's rule (Jeffrey 1965). As is usual in the field of statistical artificial intelligence, relationships that are trivial at the level of the axioms of probability are reinterpreted procedurally to give *heuristics* for belief update. We begin with Jeffrey's rule, also sometimes called *probability kinematics* (Jeffrey 1965):

$$P^{new}(H) = \sum_A P^{old}(H | A) P^{new}(A).$$

The assertion that $P^{new}(H|A) = P^{old}(H|A)$ is necessary for Jeffrey's rule to be strictly correct, but here we use it as an *Ansatz* in order to perform what Diaconis and Zabell (1982) call mechanical updating. Here “new” means “when presented with the evidence.” We interpret $P^{new}(H)$ as the revised belief given the evidence, and interpret $P^{new}(A)$ as the probability of the attribute given this evidence, and so we rewrite Jeffrey's rule as $P(H|E) = \sum_A P(H|A)P(A|E)$, using our independence assumption $P(H|A,E) = P(H|A)$. Then we have

$$\begin{aligned}
p(h_k|E) &= \sum_{i=1}^m p(h_k | a_i) p(a_i | E) \\
&= \sum_{i=1}^m p(a_i | h_k) \frac{p(h_k)}{p(a_i)} p(a_i | E) \\
&= p(h_k) \sum_{i=1}^m p(a_i | h_k) \frac{p(a_i | E)}{p(a_i)} \\
&= p(h_k) \sum_{i=1}^m p(a_i | h_k) \frac{p(a_i | E)}{\sum_{j=1}^n p(a_i | h_j) p(h_j)} .
\end{aligned}$$

The first step above uses Jeffrey's rule, the second step uses Bayes' rule, so we call this *Jeffrey-Bayes updating*. This superficially innocuous chain of calculations in fact is fraught with presuppositions. In particular, notice how the denominator “locks in” the role of the knowledge base $\{p(a_i | h_k)\}$ rather than letting it be updated by evidence; only $p(h_k)$ changes when a piece of evidence E arrives.

The final move is to take this heuristic update and make it iterative and cumulative, yielding a dynamical system. The vector $\mathbf{p}(t)$ was defined at the beginning of this section as the degree of belief in the hypothesis at time t , given the stream of evidence up through that time. Its components are

$$p_k(t) \equiv p(h_k | E(1), \dots, E(t-1)). \quad (2.1)$$

The stream of evidence comes in as

$$e_i(t) \equiv p(a_i | E(1), \dots, E(t)). \quad (2.2)$$

(“Experience” may be a more apt name than “evidence” for e_i , as it suggests the accretive nature of this dynamical process.) The rule now becomes

$$p_k(t+1) = p_k(t) \frac{\sum_{i=1}^m w_{ik} \frac{e_i(t+1)}{\sum_{j=1}^n w_{ij} p_j(t)}}{\sum_{j=1}^n w_{ij} p_j(t)} .$$

Switching to matrix-vector notation, this takes on a simple form, which we can call *Jeffrey-Bayes dynamics*:

$$\mathbf{p}(t+1) = \mathbf{p}(t) * \mathbf{W}^T \left[\frac{\mathbf{e}(t+1)}{\mathbf{W}\mathbf{p}(t)} \right]. \quad (2.3)$$

Here “*”-multiplication and “/”-division are component-wise (Hadamard) operations. The dynamics can be compactly summarized in the following pseudocode fragment (borrowing the “*” operator from the C language):

```
Initialize p;  
While new evidence e do  
    p *= WT ( e / Wp ) .
```

The vector **p** remains normalized at each step:

$$\|\mathbf{p}_{next}\|_1 = \mathbf{1}^T[\mathbf{p} * \mathbf{W}^T(\mathbf{e} / \mathbf{Wp})] = \mathbf{p}^T \mathbf{W}^T(\mathbf{e} / \mathbf{Wp}) = (\mathbf{Wp})^T(\mathbf{e} / \mathbf{Wp}) = \mathbf{1}^T \mathbf{e} = 1,$$

where **1** is the vector of all 1s, and we have used the general identities $\mathbf{1}^T(\mathbf{x} * \mathbf{y}) = \mathbf{x}^T \mathbf{y}$ and $\mathbf{x}^T(\mathbf{y} / \mathbf{x}) = \mathbf{1}^T \mathbf{y}$. Note that this particular result does not actually require that **p** be initially normalized.

If we explode the update (2.3) into four steps, its meaning becomes more clear:

$$\mathbf{a} \equiv \mathbf{Wp} \quad (2.4a)$$

$$\mathbf{s} \equiv \mathbf{e} / \mathbf{a} \quad (2.4b)$$

$$\mathbf{u} \equiv \mathbf{W}^T \mathbf{s} \quad (2.4c)$$

$$\mathbf{p}_{new} = \mathbf{p} * \mathbf{u} \quad (2.4d)$$

There are two spaces involved here. We have belief space, the simplex Δ^{n-1} where the belief vector **p** and the update vector **u** find their home. There is also observation space, the simplex Δ^{m-1} where the evidence vector **e**, the vector **s** (in effect the “surprise” content of the evidence), and the attribute vector **a** are located. The knowledge base matrix **W** is a linear transformation from belief space to observation space, and its transpose **W**^T is its dual. The dynamics is on belief space, but it is forced by external signals in observation space; see Figure 1A.

Jeffrey-Bayes dynamics is *corroborative*, in that repeated observations of the same evidence vector continually change the state. It does not *commute in evidence*,

in that the evidence stream $(\mathbf{e}, \mathbf{e}')$ drives the system to a different state in general than does $(\mathbf{e}', \mathbf{e})$. Further, if an evidence vector $\mathbf{e}$ sits at a corner of the simplex (e.g. $\mathbf{e} \sim [0 \dots 0 \ 1 \ 0 \dots 0]^T$), then the evidence is certain and there is no need for Jeffrey's probability kinematics. In this case, the product $\mathbf{W}^T (\mathbf{e} / \mathbf{W}\mathbf{p})$ extracts one row from the $\mathbf{W}$ matrix and uses $\mathbf{W}\mathbf{p}$ merely to normalize the result. Jeffrey-Bayes dynamics reduces to Bayes' rule: $p(h_k | a_i) = \alpha p(h_k) p(a_i | h_k)$, where i is the index of the one nonzero entry in $\mathbf{e}$.

The operations involved in the belief update are local, and by referring to equations (2.3) we can implement it in a connectionist fashion. We form a layer of n deterministic real-valued units to store the current belief $\mathbf{p}(t)$ and the belief update factor $\mathbf{u}(t)$. We form another layer of m units to store the attribute vector $\mathbf{a}(t)$ and the evidence vector $\mathbf{e}(t)$. The matrix $\mathbf{W}$ is the weight matrix of the network. A forward pass through the network re-expresses hypothesis probabilities as attribute probabilities, using the knowledge base. Beliefs propagate forward, surprise propagates backward. Figure 1B shows the overall architecture. Note that normalization is maintained throughout merely by means of these connectionist operations.

3 Belief Trajectories and Riemannian Geometry

Jeffrey-Bayes dynamics is motivated by some simple heuristics and has a natural connectionist implementation. A key question is whether the trajectories of such a system have a natural characterization.

If $\mathbf{e}$ is in the image of Δ^{n-1} under $\mathbf{W}$ then we can have $\mathbf{W}\mathbf{p}=\mathbf{e}$. For this value of $\mathbf{p}$ the dynamics is at a stationary point: $\mathbf{W}^T(\mathbf{e}/\mathbf{W}\mathbf{p}) = \mathbf{W}^T \mathbf{1} = \mathbf{1}$ (where $\mathbf{1}$ is the vector of all 1s), here using the fact that the columns of $\mathbf{W}$ sum to 1. In general, the equation $\mathbf{W}\mathbf{p}=\mathbf{e}$ for arbitrary $\mathbf{W}$ and $\mathbf{e}$ will have no solution on the probability simplex. Nevertheless we shall see that the trajectories change $\mathbf{p}$ to make $\mathbf{W}\mathbf{p}$ closer to $\mathbf{e}$.

A natural measure of closeness on a probability simplex is the Kullback-Leibler divergence:

$$D(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^n p_i \log \frac{p_i}{q_i}$$

In studying this system we monitor the quantity $f(\mathbf{p}) = D(\mathbf{e}, \mathbf{W}\mathbf{p})$ for fixed $\mathbf{W}$ and $\mathbf{e}$. Figures 2 and 3 show contour plots of $f(\mathbf{p})$ on Δ^2 for particular $\mathbf{W}$ and $\mathbf{e}$ choices, and trace out the trajectories of the Jeffrey-Bayes dynamics starting from two initial states with the evidence clamped at a constant value $\mathbf{e}(t) = \mathbf{e}_0$. All trajectories come to rest at a minimum of $f(\mathbf{p})$, corresponding to a belief state which is maximally harmonized with the evidence and knowledge base. However, the trajectories do not follow the

usual paths of steepest descent, also shown in the figure. Indeed, this neural network seems to be somewhat indecisive in its reluctant S-shaped swerves toward equilibrium!

Jeffrey's rule on its own updates a probability distribution P to pick out, from all the distributions consistent with the evidence, the one that minimizes Kullback-Leibler divergence from P (Diaconis and Zabell 1982). The Jeffrey-Bayes rule is more complex. As it turns out, it *is* doing steepest descent in Kullback-Leibler divergence, as long as we get the geometry of the space right. Following the practice of information geometry (Amari 2000), we view belief space as a Riemannian space endowed with a metric tensor given by the Fisher information. In general, with coordinates $\boldsymbol{\eta} = (\eta^1, \eta^2, \dots, \eta^{n-1})$ an information manifold of probability distributions $\{p_{\boldsymbol{\eta}}(x)\}$ has the natural metric

$$g_{ij}(\boldsymbol{\eta}) = \int \left[\frac{\partial}{\partial \eta^i} \log p_{\boldsymbol{\eta}}(x) \frac{\partial}{\partial \eta^j} \log p_{\boldsymbol{\eta}}(x) \right] p_{\boldsymbol{\eta}}(x) dx$$

For the case of the space Δ^{n-1} , the variable x becomes the index k , and we identify the parameter of the probability density with the probabilities themselves: $p_{\boldsymbol{\eta}}(k) = p_k$. The $n-1$ coordinates $\boldsymbol{\eta}$ completely determine the probability vector $\mathbf{p}$, since

$$p_n = 1 - \sum_{k=1}^{n-1} \eta^k. \quad (3.1)$$

Accordingly,

$$g_{ij}(\boldsymbol{\eta}) = \sum_{k=1}^n \left[\frac{\partial}{\partial \eta^i} \log p_k \frac{\partial}{\partial \eta^j} \log p_k \right] p_k = \frac{\delta_{ij}}{\eta^i} + \frac{1}{p_n} \quad (3.2)$$

The value of this metric varies across the manifold. In fact it is related to the Hessian of the absolute entropy:

$$g_{ij} = -\frac{\partial^2 H}{\partial \eta^i \partial \eta^j} \quad \text{where } H = -\sum_{k=1}^n p_k \log p_k.$$

The inverse of this matrix $\mathbf{G}(\boldsymbol{\eta}) = (g_{ij})$ is the matrix $\mathbf{G}^{-1}(\boldsymbol{\eta})$ with components

$$g^{ij} = \eta^i \delta_{ij} - \eta^i \eta^j \quad (3.3)$$

or, in matrix form, $\mathbf{G}^{-1}(\boldsymbol{\eta}) = \text{diag } \boldsymbol{\eta} - \boldsymbol{\eta} \boldsymbol{\eta}^T$.

A *natural* gradient steepest descent algorithm (Amari 1998) modifies the gradient using the metric tensor:

$$\boldsymbol{\eta} \rightarrow \boldsymbol{\eta} - \varepsilon \mathbf{G}^{-1}(\boldsymbol{\eta}) \nabla_{\boldsymbol{\eta}} \phi(\boldsymbol{\eta}) \quad (3.4)$$

If we had viewed Δ^{n-1} as a Euclidean space, we would have been tempted to use $\boldsymbol{\eta} - \varepsilon \nabla_{\boldsymbol{\eta}} \phi(\boldsymbol{\eta})$. This gives the curves in Figure 2 that appear to be steepest descent curves, in that they seem to be orthogonal to the contours. But they are *not* orthogonal given the appropriate metric (3.2). One can view the introduction of the factor $\mathbf{G}^{-1}$ as the rectification of a simple “type mismatch.” Whereas $\boldsymbol{\eta}$ is a vector, the gradient $\nabla_{\boldsymbol{\eta}} \phi$ is technically a covector, and these distinct types cannot be added together. Multiplication by the inverse of the metric tensor performs the appropriate conversion to a vector, thereby allowing the addition operation.

We now show that one step of the Jeffrey-Bayes dynamics is a unit step ($\varepsilon=1$) in natural gradient descent using the Fisher metric. In the derivation that follows we will make use of the abbreviations $\mathbf{s}$ and $\mathbf{a}$ as defined in (2.4), and view $\mathbf{s}$, $\mathbf{a}$ and $\mathbf{p}$ all as functions of the coordinates $\boldsymbol{\eta}$. The objective function is

$$\phi(\boldsymbol{\eta}) = D(\mathbf{e}, \mathbf{W}\mathbf{p}) = \sum_{i=1}^m e_i \log \frac{e_i}{a_i} = -H(\mathbf{e}) - \sum_{i=1}^m e_i \log a_i$$

where $H(\mathbf{e}) = -\sum_i e_i \log e_i$ is the entropy of the evidence vector.

The derivatives evaluate to:

$$\frac{\partial \phi}{\partial \eta^k} = -\sum_{i=1}^m e_i \frac{\partial}{\partial \eta^k} \log a_i = -\sum_{i=1}^m \frac{e_i}{a_i} \frac{\partial a_i}{\partial \eta^k}$$

Since $\mathbf{s}=\mathbf{e}/\mathbf{a}$ and $\mathbf{a}=\mathbf{W}\mathbf{p}$, using (3.1) yields

$$\frac{\partial \phi}{\partial \eta^k} = -\sum_{i=1}^m s_i \frac{\partial}{\partial \eta^k} \left[\sum_{j=1}^{n-1} w_{ij} \eta^j + w_{in} \left(1 - \sum_{r=1}^{n-1} \eta^r \right) \right] = -\sum_{i=1}^m s_i (w_{ik} - w_{in}) \quad (3.5)$$

This gives us the spatial derivative of ϕ at every point on the belief simplex. Taking $\varepsilon=1$, steepest descent in the natural gradient will update η^r to $\eta^r - \gamma^r$, where

$$\gamma^r \equiv [\mathbf{G}^{-1} \nabla_{\boldsymbol{\eta}} \phi(\boldsymbol{\eta})]^r = \sum_{k=1}^{n-1} g^{rk} \frac{\partial \phi}{\partial \eta^k}. \quad (3.6)$$

From (3.1,3,5,6), we obtain the updated value of η^r as

$$\begin{aligned}
\eta^r - \gamma^r &= \eta^r - \sum_{k=1}^{n-1} (\eta^r \delta_{rk} - \eta^r \eta^k) \left[- \sum_{i=1}^m s_i (w_{ik} - w_{in}) \right] \\
&= \eta^r \left[1 - \sum_{i=1}^m w_{in} s_i + \sum_{k=1}^{n-1} \eta^k \sum_{i=1}^m w_{in} s_i - \sum_{k=1}^{n-1} \eta^k \sum_{i=1}^m w_{ik} s_i + \sum_{i=1}^m w_{ir} s_i \right] \\
&= \eta^r \left[1 - p_n \sum_{i=1}^m w_{in} s_i - \sum_{k=1}^{n-1} \eta^k \sum_{i=1}^m w_{ik} s_i + \sum_{i=1}^m w_{ir} s_i \right] \\
&= \eta^r \left[1 - \sum_{k=1}^n p_k \sum_{i=1}^m w_{ik} s_i + \sum_{i=1}^m w_{ir} s_i \right] \\
&= \eta^r \left[1 - \sum_{i=1}^m s_i \sum_{k=1}^n w_{ik} p_k + \sum_{i=1}^m w_{ir} s_i \right] \\
&= \eta^r \left[1 - \sum_{i=1}^m e_i + \sum_{i=1}^m w_{ir} s_i \right] \\
&= \eta^r \left[\sum_{i=1}^m w_{ir} s_i \right].
\end{aligned}$$

Therefore the natural gradient steepest descent will update $\mathbf{p}$ by the factor $\mathbf{W}^T \mathbf{s} = \mathbf{W}^T (\mathbf{e}/\mathbf{W}\mathbf{p})$, identical to what we produced by consideration of Bayes' and Jeffrey's rules. Notice how an additive update (subtraction of the gradient) has become a multiplicative update without any explicit exponentiation. This result shows that the network's seemingly indecisive swerves are indeed "natural;" the Fisher metric tensor is built in to this connectionist architecture.

4 Discussion

That such a simple form of Jeffrey-Bayes updating is equivalent to natural steepest descent in $D(\mathbf{e}, \mathbf{W}\mathbf{p})$ – with unit step size no less – suggests that this truly is a basic mode of probabilistic inference. (Steepest descent using the natural gradient of $D(\mathbf{W}\mathbf{p}, \mathbf{e})$ is readily derivable and yields the same asymptotic behavior, but does not admit such a simple form.) Is this geometrically natural connectionist computation

also *biologically* natural? A full study of biologically realistic implementations is beyond the scope of this paper, but some aspects are worth mentioning.

First, for better or worse, the backward propagation in the Jeffrey-Bayes algorithm (i.e., multiplication by $\mathbf{W}^T$) makes implementational demands similar to classic backpropagation. Instead of backpropagating error vectors, however, we backpropagate surprise vectors. No synaptic modification takes place.

Second, the Jeffrey-Bayes architecture shares with the single-layer cortical circuit models of Rao (2004) the incorporation of long-term prior knowledge in the connection weights, as well as the continual revision of posterior probabilities through the operation of a recurrent circuit. Both models perform Bayesian computations, although Rao’s architectures are designed to realize more powerful hidden Markov models. Both deal with posterior probabilities over time like (2.1) and (2.2). In contrast, firing frequencies in Rao’s circuits encode *log* probabilities. Taking the logarithm of the Jeffrey-Bayes update step, and using the same “log-of-sum as sum-of-logs” approximation as Rao, would bring Jeffrey-Bayes to a roughly equivalent level of basic biological plausibility. (Interestingly, Jeffrey-Bayes trajectories easily degenerate to straight lines in log space, i.e., to *e*-geodesics; see Figure 3.) However, transforming to the log domain is not as urgent in this architecture. The multiplications performed in our model do not demand a full multiplicative neural network. The component-wise operations take only two factors. The neurons storing the $\mathbf{a}$ and $\mathbf{u}$ vectors are in effect modulatory interneurons. The division by a_i could be approximated by a transfer function suitably close to $f(x)=1/x$. Further, each step in the Jeffrey-Bayes forward-backward update renormalizes the belief vectors, so even if there is some inaccuracy in the coding of $\mathbf{p}(t)$ that leads it wander off the simplex Δ^{n-1} into the broader space $\mathbf{R}^{n-1}$ it will be snapped back.

Despite the example cases illustrated in Figures 2 and 3, where initial beliefs are driven by *constant* evidence vectors to a stationary point, we view the most interesting mode of operation of this circuit to be when the evidence $\mathbf{e}(t)$ is changing slowly in time. In this case, we imagine an “adiabatic” scenario wherein the network tracks moving evidence vectors and stays in a minimum of KL divergence. In this case, the neuronal layer representing beliefs in hypotheses naturally adapts to track slowly varying vectors $\mathbf{e}(t)$. Using probability kinematics in Riemannian space to seek the equilibrium state $\mathbf{p}$ is more natural, in both neural and geometric senses, than, say, directly computing the pseudoinverse $\mathbf{W}^+$ and projecting $\mathbf{W}^+\mathbf{e}$ back on to the simplex. The weights in $\mathbf{W}$ could be regarded as innate, or, more interestingly, adapted on an even slower time scale.

Finally, the two salient features of Jeffrey-Bayes dynamics, its non-commutativity in evidence (shared by Jeffrey’s rule) and its corroborativity (not shared by Jeffrey’s rule, which is idempotent), are in some general sense

psychologically natural. If we hear the speech of presidential candidate $e(1)$ followed by candidate $e(2)$, will we end up in the same final state as we would hearing $e(2)$ followed by $e(1)$? Will listening to the $e(1)$ speech a second time leave our beliefs unchanged? Arguably no and no, especially if we use the suggestive term “experience” rather than “evidence” for $e_i(t)$ in (2.2). Because these features emerge from geometrically natural connectionist computations, Jeffrey-like rules may ultimately find their ideal home not in the philosophy of science and statistics – their birthplace, where non-commutativity is usually viewed as *unnatural* – but in the field of cognitive neuroscience.

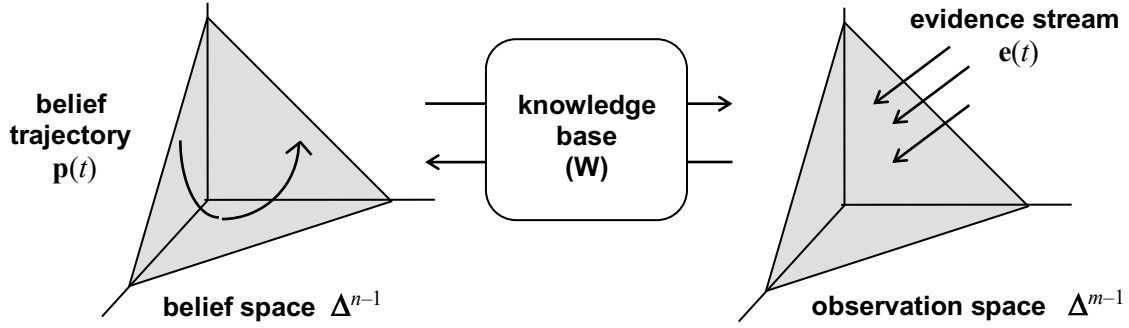
Acknowledgements

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(A)



(B)

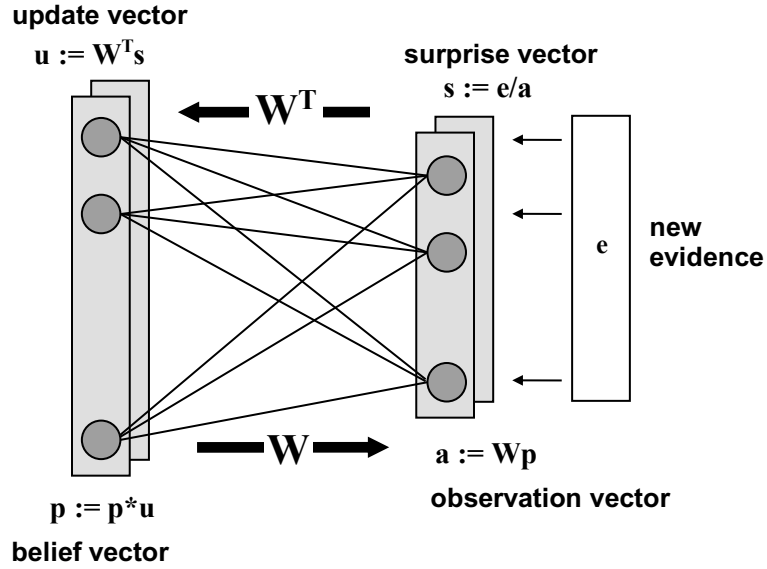


Figure 1. A schematic view of Jeffrey-Bayes dynamics (A), showing two-dimensional simplexes for belief space and observation space. Belief states propagate through the knowledge-base matrix W to observation space, yielding attribute probabilities that are adjusted by the evidence. These are propagated back to the belief space where they update the degree of belief in the hypotheses. This can be straightforwardly implemented as a two-layer recurrent network (B), where signals are passed backward and forward through the same fixed weights.

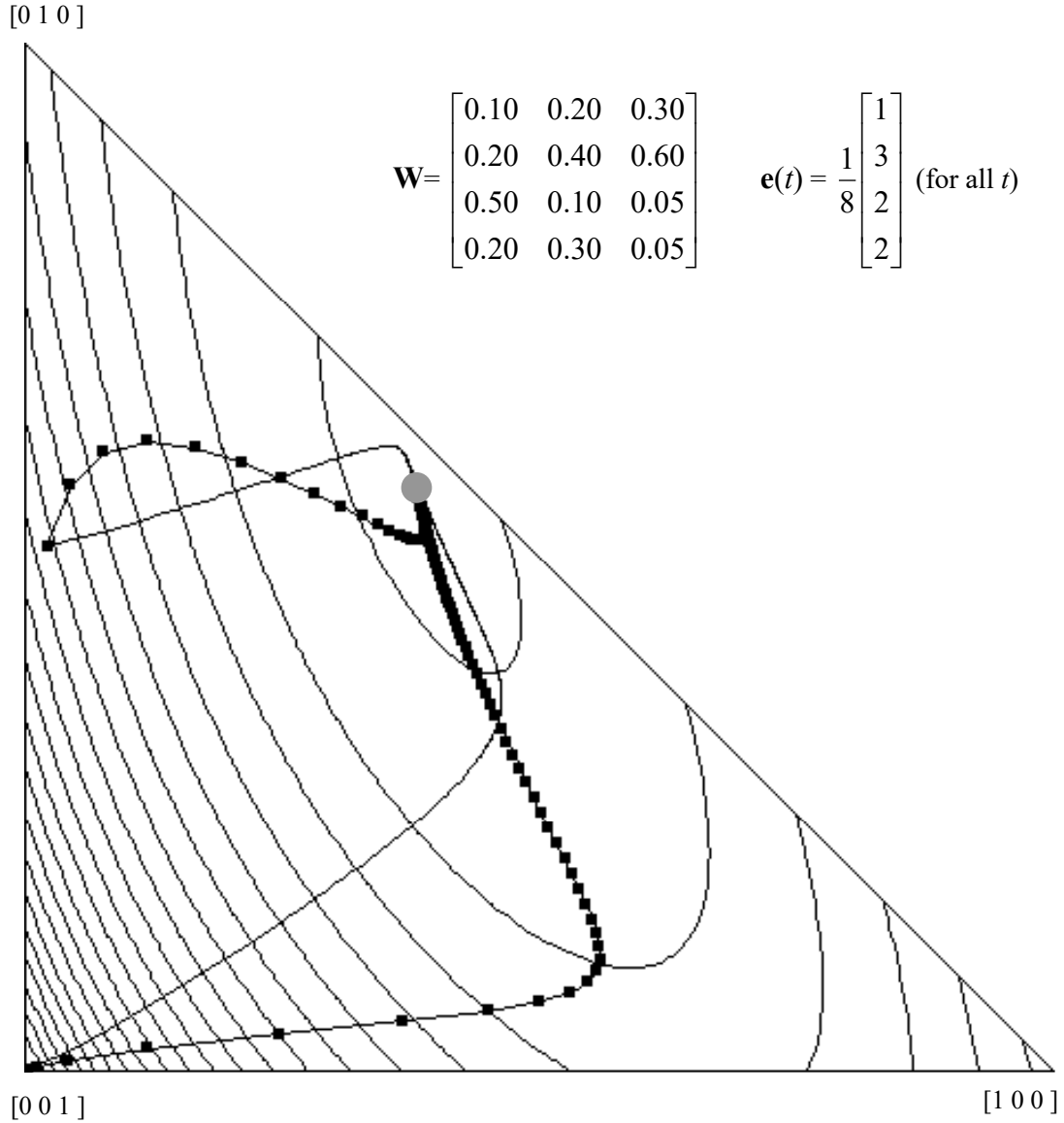


Figure 2. Example trajectories $\mathbf{p}(t)$ from two initial points on a two-dimensional simplex. The Jeffrey-Bayes dynamics (boxes) is equivalent to unit step Riemannian steepest descent in the Kullback-Leibler divergence between clamped evidence $\mathbf{e}$ and the image $\mathbf{W}\mathbf{p}$ of the belief vector $\mathbf{p}$ in observation space. The smooth curves show the paths of Euclidean steepest descent from the same two initial points. Both reach the same minimum point (gray disk), but the swerving paths of the former are geometrically more natural in the Fisher metric.

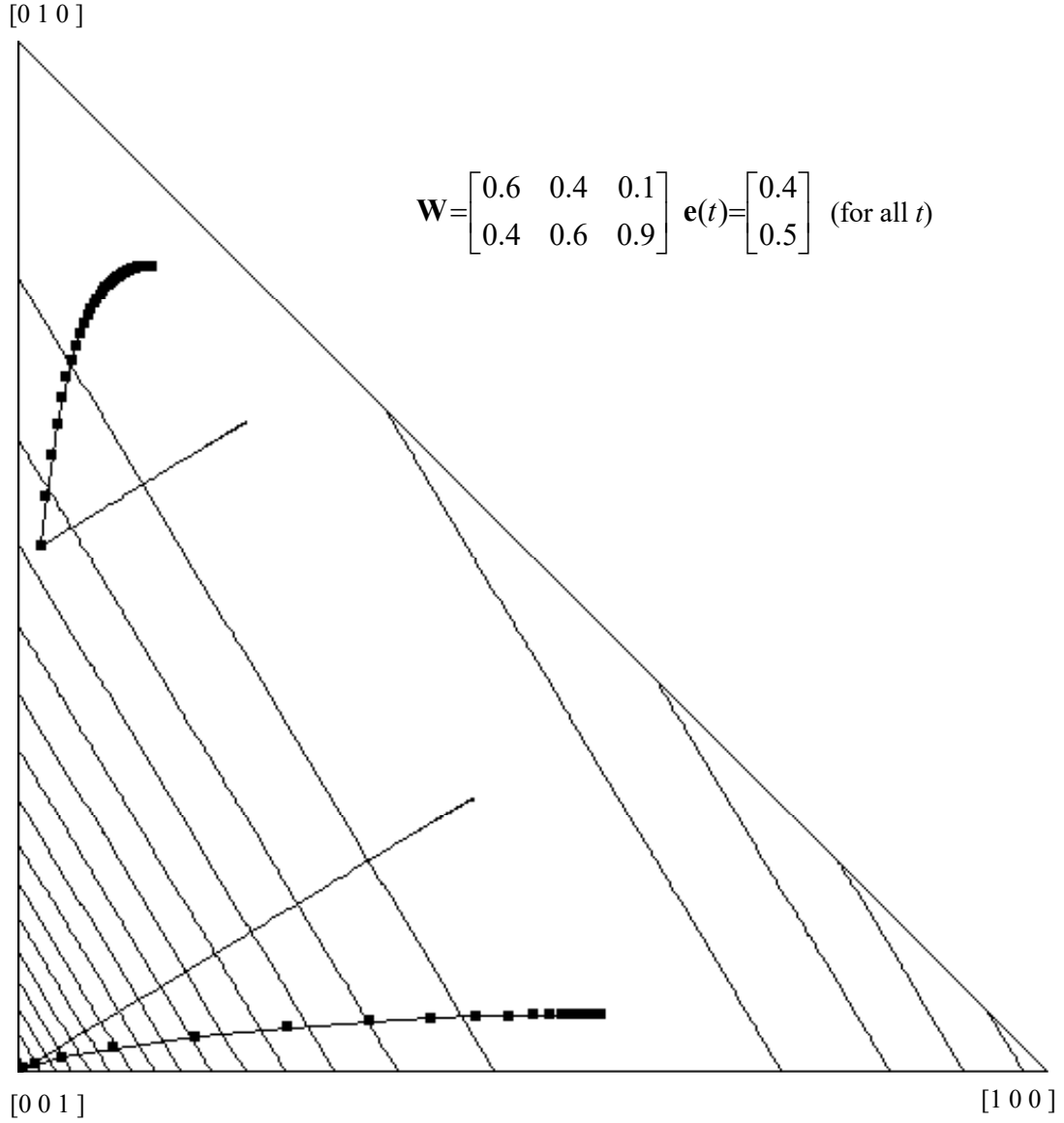


Figure 3. Belief trajectories $\mathbf{p}(t)$ from two the same two initial points used in Figure 2, but with a lower dimensional attribute space, so that the minimal values of the KL divergence are at the bottom of a linear trough. In this case, the contours are parallel, in fact on the 2D simplex they are m -geodesics (mixtures of two probabilities), as are the “straight” Euclidean steepest descent paths. The Jeffrey-Bayes trajectories appear curved, but in fact are orthogonal to these m -geodesic contours in the Fisher metric, and thus are e -geodesics (mixtures of two log probabilities).